

Simulation-based Time Series Analysis:

Likelihood-free Estimation, Testing for Structure, and Prediction with Transformers using Synthetic Data

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Overview of presentation

- ▶ Introduction to time series

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- ▶ Conditional independence testing

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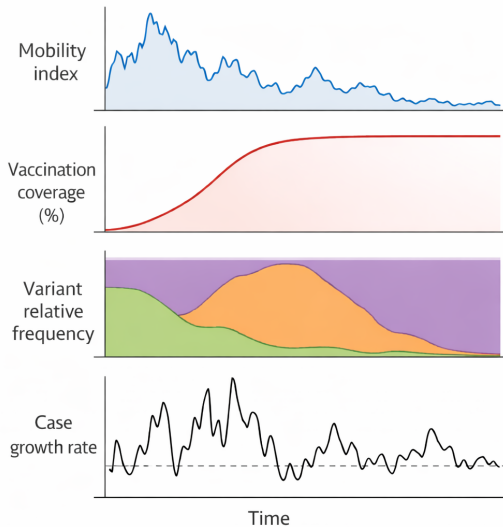
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- ▶ **Nonstationarity:** the joint distribution of $X_{t_1+\tau}, \dots, X_{t_m+\tau}$ at $m \in \mathbb{N}$ distinct time points $t_1, \dots, t_m$ may change with the time-offset τ

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- ▶ Example: time series in epidemiology

COVID-19 time series May 2020–2022 ($n=365 \times 2=730$, $d=4$)



Conditional independence testing

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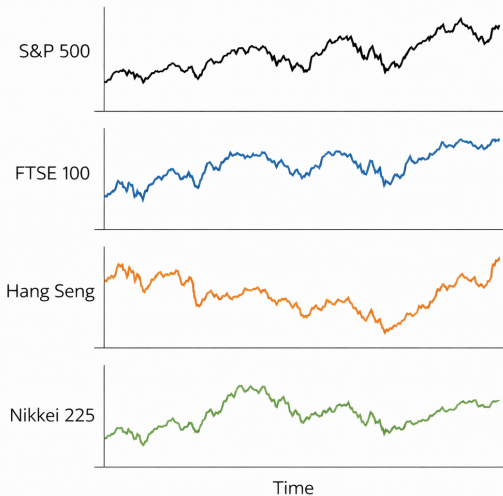
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- ▶ We show a subset of the results based on BH-adjusted p-values

Closing prices of each stock index over time



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- ▶ We reject every null hypothesis at the significance level $\alpha = 0.05$

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- ▶ Next, we present the testing procedure

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- **Inputs:** Data $X_{1:n}$, $Y_{1:n}$, $Z_{1:n}$, significance level α , number of simulations s , $p \in [2, \infty]$ for norm in test statistic, and method for selecting the window size L

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- ▶ Calculate test statistic on time series of residual products

$$T(\hat{R}_{1:n}) = \max_{j \in [n]} \left\| \frac{1}{\sqrt{n}} \sum_{t \leq j} \hat{R}_t \right\|_p$$

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- ▶ Reject null hypothesis at level α if $T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\text{boot}}$, else retain

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- ▶ **Key assumption:** uniform decay of physical dependence measure (Wu 2005)

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- ▶ Let $G_t^{(n)} : \mathbb{R}^\infty \times \Theta \rightarrow \mathbb{R}^d$ be some measurable function, $t, n, d \in \mathbb{N}$

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- ▶ Given a parameter $\theta \in \Theta$, index $n \in \mathbb{N}$ (linked to sample size and approximation), and noise inputs ϵ_t , a time series $X_{1:n}$ is *generated by nature* as

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- ▶ Denote the collections by $\mathcal{P}_n = \{P_{\theta,n} : \theta \in \Theta\}$ and $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$

Key assumption: Decaying influence of noise inputs (Wu 2005)

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- There exist $\Psi, \rho > 0$ such that, for some $q > 2$ and all $j \in \mathbb{N}$, we have

$$\sup_{\theta \in \Theta} \sup_{\ell \in \mathbb{Z}} \sup_{i \in \mathbb{N}} \sup_{t \in \mathbb{N}} \left\| G_t^{(i)}(\epsilon_\ell, \theta) - G_t^{(i)}(\tilde{\epsilon}_{\ell,j}, \theta) \right\|_{L^q(\theta)} \leq \Psi(j \vee 1)^{-\rho}$$

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where $\|\cdot\|_{L^q(\theta)} = \mathbb{E}_\theta (\|\cdot\|_2^q)^{1/q}$, $\mathbb{E}_\theta(\cdot)$ is expectation w.r.t. θ -dependent distribution

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where $\|\cdot\|_{L^q(\theta)} = \mathbb{E}_\theta (\|\cdot\|_2^q)^{1/q}$, $\mathbb{E}_\theta(\cdot)$ is expectation w.r.t. θ -dependent distribution

- ▶ We make this assumption for $X_{1:n}$, $Y_{1:n}$, and $Z_{1:n}$

We prove our test has uniformly asymptotic Type I error control

- For each $n \in \mathbb{N}$, let $\mathcal{P}_n^* \subset \mathcal{P}_n^{\text{CI}}$ be a collection of joint distributions for $X_{1:n}$, $Y_{1:n}$, and $Z_{1:n}$ for which the null hypothesis holds *and all assumptions are satisfied*

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Theorem (Informal)

Recall the test statistic $T(\hat{R}_{1:n})$ and estimated quantile $\hat{q}_{1-\alpha}^{\text{boot}}$. We have

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n^*} \mathbb{P}_P \left(T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\text{boot}} \right) \leq \alpha.$$

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- ▶ **Key assumption:** We know $G_t^{(n)}$ and can simulate from it at any value of $\theta \in \Theta$

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- ▶ Automatic feature selection, computationally efficient, little knowledge needed

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- ▶ If physical dependence measure (Wu 2005) decays uniformly over Θ , we can estimate $\Phi(\theta)$, $\theta \in \Theta$, via time-averages of $2p + 1$ random features

$$\frac{1}{n-m} \sum_{t=m+1}^n \varphi \left([X_t(\theta), \dots, X_{t-m}(\theta)]^\top \right)$$

Example: Mean of Gaussian

- **Goal:** Estimate $\theta_0 = 0$ given $X_1^{\text{obs}}, \dots, X_n^{\text{obs}} \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta_0, 1)$ with $\Theta = [-3, 3]$

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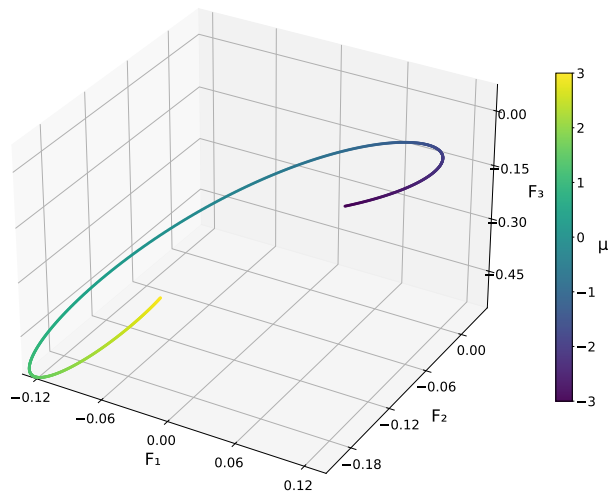
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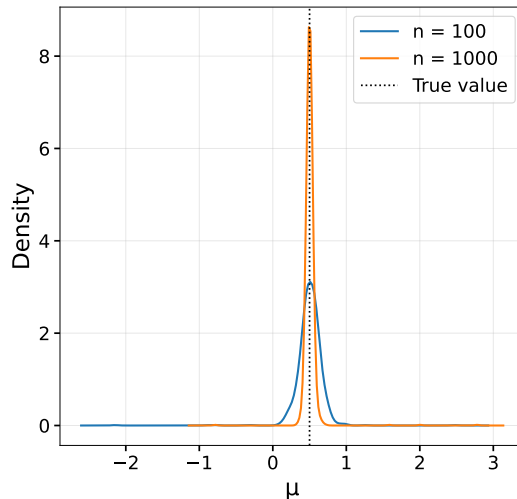
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- ▶ We will compare our random feature estimator $\hat{\theta}$ with $\hat{\theta}^{\text{MLE}} = \frac{1}{n} \sum_{t=1}^n X_t$

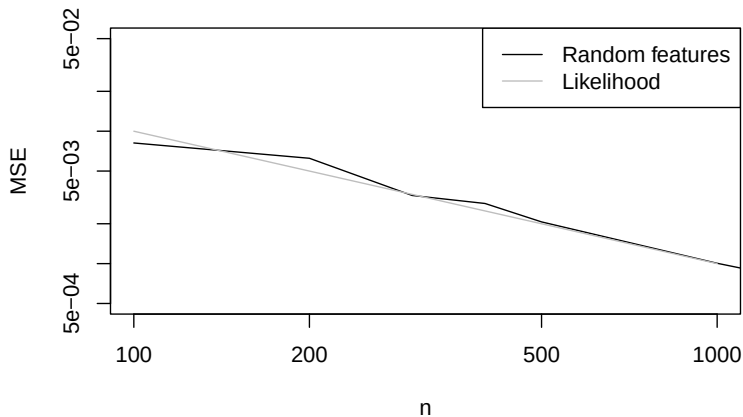
Time-average of 3 features, colored by parameter value ($n = 1000$, $s = 10$)



Density plot based on 1,000 independent estimates



Theoretical MSE of MLE vs MSE of our estimator (100 replicates per n)



Theoretical guarantees for random feature estimators

- We prove consistency $\hat{\theta} \xrightarrow{P} \theta_0$ and asymptotic normality

$$\sqrt{n} \left(\hat{\theta} - \theta_0 \right) \Rightarrow N \left(0, (\Gamma^\top \Gamma)^{-1} \Gamma^\top V \Gamma (\Gamma^\top \Gamma)^{-1} \right)$$

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- ▶ Rolling-window estimator also introduced, and we prove analogous guarantees using the framework of locally stationary time series
- ▶ Examples: ODEs observed through dependent noise, nonstationary state-space models, and structural time series models

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- ▶ We develop theory, and propose methods for smoothing and forecasting

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- ▶ For forecasting with TSFMs, Dooley et al. (2023) suggest training on simulations from multiplicative seasonality-trend-noise decomposition models (many $\theta \in \Theta$)
- ▶ Consider a version with multiplicative autoregressive noise

$$Y_t = \text{Se} \left(t, \theta^{\text{Se}} \right) \text{Tr} \left(t, \theta^{\text{Tr}} \right) \exp \left(\kappa X_t - \frac{\kappa^2 \sigma_X^2}{2(1 - \phi^2)} \right) \exp \left(\varepsilon_t^Y - \frac{\sigma_Y^2}{2} \right)$$
$$X_t = \phi X_{t-1} + \varepsilon_t^X$$

for some scaling factor $\kappa > 0$ and autoregressive coefficient $\phi \in (-1, 1)$, with $X_0 \sim N(0, \frac{\sigma_X^2}{1-\phi^2})$ and $\varepsilon_t^X \stackrel{\text{iid}}{\sim} N(0, \sigma_X^2)$, $\varepsilon_t^Y \stackrel{\text{iid}}{\sim} N(0, \sigma_Y^2)$ for some $\sigma_X, \sigma_Y > 0$

- ▶ Seasonality: linear combination of sines, cosines (weekly, monthly, yearly periods)
- ▶ Trend: product of linear trend and exponential trend
- ▶ Our rolling-window estimator can be used (consistent and asymptotically normal)

Unifying framework: Simulation-and-regression with parameter uncertainty

- We propose minimizing the Monte Carlo approximation to

$$R_{P_{\widehat{\text{mix}}}}(f) = \mathbb{E}_{P_{\widehat{\text{mix}}}} \left[\sum_{t \in \mathcal{T}} L(X_t - [f(Y)]_t) \right] = \mathbb{E}_{\theta \sim \widehat{\text{CD}}_n} \left[\mathbb{E}_{P_\theta} \left[\sum_{t \in \mathcal{T}} L(X_t - [f(Y)]_t) \right] \right]$$

for some loss L , where the expectation is w.r.t. the mixture distribution

$P_{\widehat{\text{mix}}} = \int_{\Theta} P_\theta d\widehat{\text{CD}}_n(\theta)$ of sequences X, Y corresponding to some *estimated confidence distribution* $\widehat{\text{CD}}_n$ over $\Theta \subset \mathbb{R}^p$ for θ_0 (D. Liu, R. Y. Liu, and Xie 2022)

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- When $\widehat{\text{CD}}_n$ is uniform over Θ , this is like TSFM pre-training (no information)
- When $\widehat{\text{CD}}_n$ is Dirac delta at $\hat{\theta}$, this is like classical simulation-and-regression, which aims to minimize the Monte Carlo approximation to

$$R_{P_{\hat{\theta}}}(f) = \mathbb{E}_{P_{\hat{\theta}}} \left[\sum_{t \in \mathcal{T}} L(X_t - [f(Y)]_t) \right]$$

where expectation is w.r.t. distribution $P_{\hat{\theta}}$ of sequences X, Y corresponding to $\hat{\theta}$

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- ▶ **Calibration:** Does accounting for parameter uncertainty improve the calibration of pre-trained/untrained transformers that output multiple conditional quantiles?

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- ▶ **Calibration:** Does accounting for parameter uncertainty improve the calibration of pre-trained/untrained transformers that output multiple conditional quantiles?
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- ▶ **Fine-tuning and simulation-and-regression:** When does accounting for parameter uncertainty improve on simulation-and-regression methods and TSFM fine-tuning methods that are based on parameter point estimates?

Thank you!

Questions or comments?

You can reach me at:
`mwiecksosa@cmu.edu`

Does accounting for parameter uncertainty improve the calibration of pre-trained/untrained transformers that output conditional quantiles?

- ▶ For untrained transformers, see answer to question: “In the context of simulation-and-regression, does accounting for parameter uncertainty achieve better performance than using parameter estimate?”

Does accounting for parameter uncertainty improve the calibration of pre-trained/untrained transformers that output conditional quantiles?

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In the context of simulation-and-regression, does accounting for parameter uncertainty achieve better performance than using parameter estimate?

- When does this inequality hold:

$$\mathbb{E}_{P_{\theta_0}} \left[\sum_{t \in \mathcal{T}} L \left(X_t - [f_{\widehat{P}_{\text{mix}}}^*(Y)]_t \right) \right] < \mathbb{E}_{P_{\theta_0}} \left[\sum_{t \in \mathcal{T}} L \left(X_t - [f_{P_{\hat{\theta}}}^*(Y)]_t \right) \right]$$

where both expectations are with respect to the distribution P_{θ_0} of the sequences corresponding to the unknown true parameter θ_0 , $f_{\widehat{P}_{\text{mix}}}^*$ is the minimizer in the function class of $R_{\widehat{P}_{\text{mix}}}(f)$, and $f_{P_{\hat{\theta}}}^*$ is the minimizer in the function class of $R_{P_{\hat{\theta}}}(f)$

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- ▶ Intuitively, this occurs when the following conditions hold
- ▶ Large parameter uncertainty, e.g., when the sample size of the observed time series is small relative to the noise
- ▶ The distributions used in the pre-training phase are close to the true distribution, e.g., when we have domain knowledge or we have previously observed time series with similar distributions

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